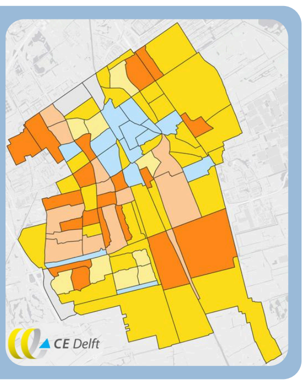
The Delft Heat Transition

Environmental Impact of Infrastructure

Background and context

- Delft Heat Plan: gas-free by 2050
- Transition to sustainable heating technologies
- Previous work on cost of heating technologies per neighborhood, qualitative assessment of local environmental impacts

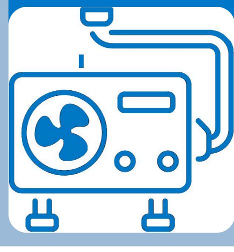


Research question

What are the environmental impacts of the technologies proposed for the Delft heat transition?

Scenario 1: District heating

Scenario 2: Electrification



Scenario 3: Hybrid

Methodology

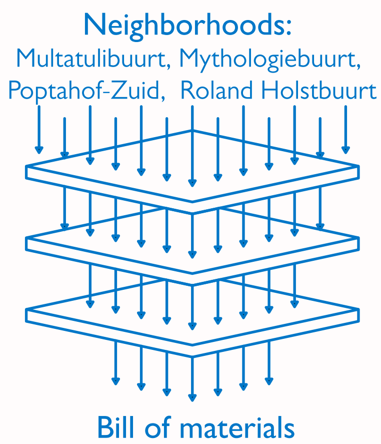
I. Life cycle assessment

Goal and scope definition → Inventory analysis → Impact assessment → Interpretation

Output: environmental impacts per functional unit

II. System modeling

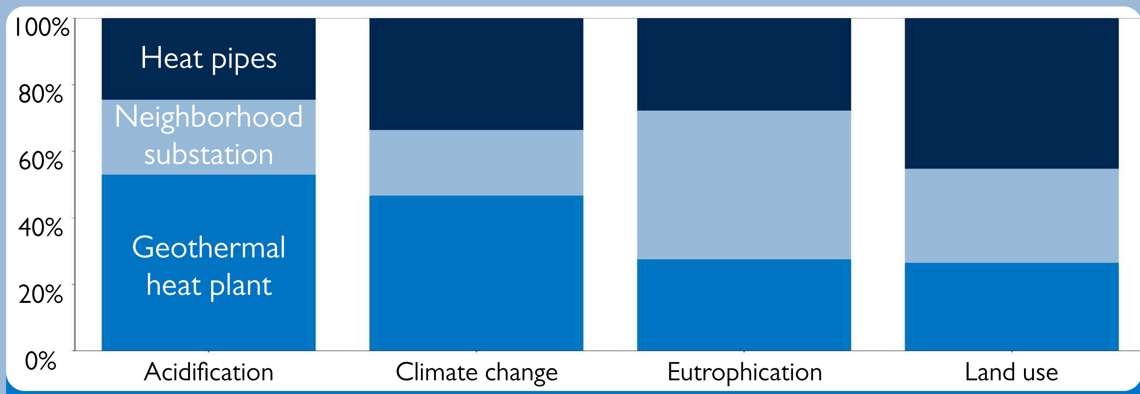
1. Input neighborhood heat demand and spatial data
2. Define network nodes and links
3. Optimize the system for minimum capacity



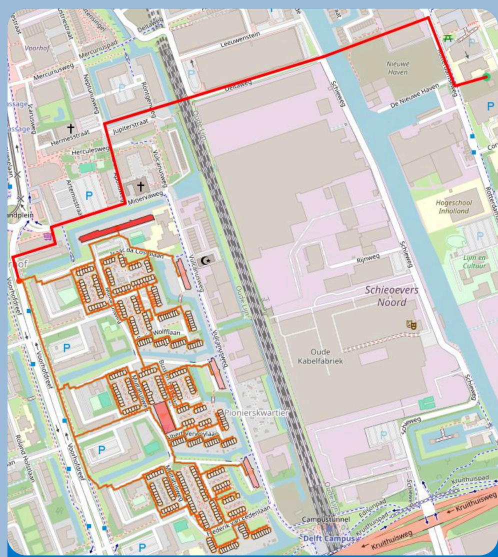
III. Integrated scenario analysis

Bill of materials → Calculate impacts: Scenario 1, Scenario 2, Scenario 3 → Compare impacts: Neighborhoods, Pathways

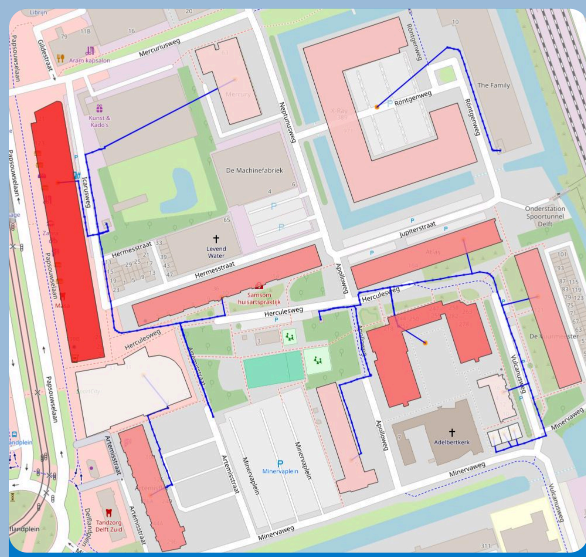
Results



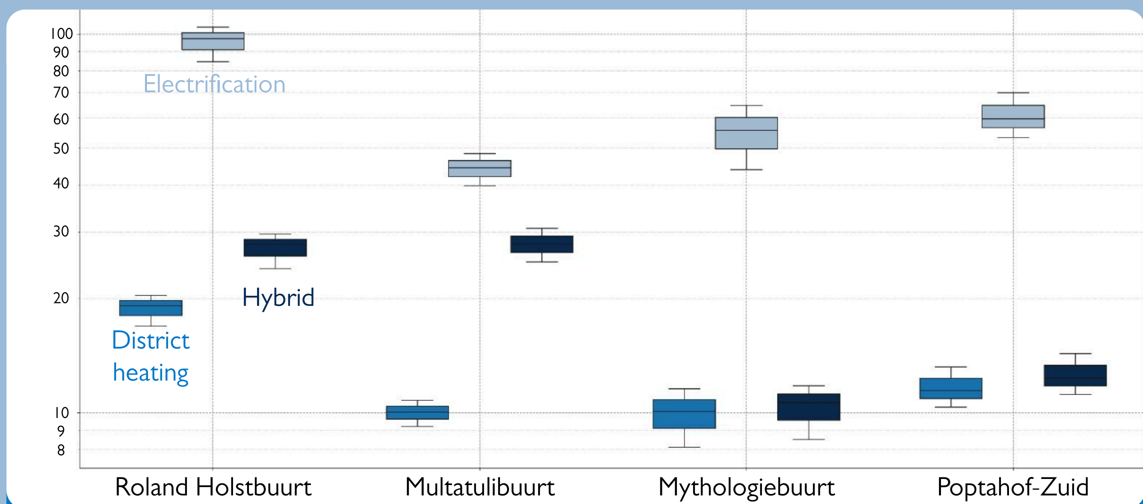
Per-component contributions to district heating environmental impacts, Multatulibuurt



District heating model, Multatulibuurt



Electrification model, Mythologiebuurt



Climate change impacts in ktons CO_2-eq, for each scenario and neighborhood

Conclusions

1. Lowest impact option differs depending on impact category
2. District heating scenario demonstrates lower CO_2 emissions than electrification or hybrid scenarios
3. Hotspots: steel manufacturing (geothermal plant, substation, piping), cement, refrigerant (air-to-air heat pumps)
4. Few differences between neighborhoods despite differing layouts and demand structure

Limitations and future work

- Alternative heating technologies (waste heat, storage) and upstream systems (additional electricity generation, MV/HV grid) not modeled
- Operational impacts not evaluated - ratio of operational to embedded impacts only extrapolated from literature
- Only four neighborhoods studied, but results are robust between cases
- Sensitivity to system parameters currently under study

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